

Numerical Analysis PhD Exam, January 2000

Do any 8 of the following 10 problems.

1. This problem has four parts.

- (1) Show: If a matrix $A \in \mathbb{R}^{n \times n}$ has eigenvalues $\lambda_1, \dots, \lambda_n$ such that $\lambda_i \neq \lambda_j$ for all $i \neq j$, then the corresponding eigenvectors are linearly independent.
- (2) Show: If a symmetric matrix $A \in \mathbb{R}^{n \times n}$ has eigenvalues $\lambda_1, \dots, \lambda_n$ such that $\lambda_i \neq \lambda_j$ if $i \neq j$, then the eigenvectors of A are orthogonal.
- (3) Show: If the matrix $A \in \mathbb{R}^{n \times n}$ has n independent eigenvectors, then the matrix can be diagonalized.
- (4) Show: If $A \in \mathbb{R}^{n \times n}$ is symmetric, then the matrix can be diagonalized by an orthogonal matrix.

2. This problem has three parts.

- (1) Give a careful statement of the singular value decomposition for an arbitrary matrix $A \in \mathbb{R}^{m \times n}$.
- (2) Give a careful proof of the singular value decomposition.
- (3) Show how the singular value decomposition for A naturally yields a solution to the least squares problem $\min_x \|Ax - y\|_2$.

3. Warning: Suspected error. The source calls H an “orthogonal projection,” but $H = I - 2ww^T$ is generally a Householder reflection (and an orthogonal matrix), not a projection. The source wording was retained because the intended replacement is not uniquely determined.

This problem has three parts.

- (1) Show: If w is a unit vector in $\mathbb{R}^n$, then the matrix $H = I - 2ww^T$ is an orthogonal projection.
- (2) Show: If $w = \frac{x-y}{\|x-y\|}$, $\|x\| = \|y\|$, and $H = I - 2ww^T$, then $Hx = y$.
- (3) Given an arbitrary matrix $A \in \mathbb{R}^{m \times n}$, construct an orthogonal matrix $Q \in \mathbb{R}^{m \times m}$ and an upper triangular matrix $R \in \mathbb{R}^{m \times n}$ such that $A = QR$.

4. This problem has three parts.

- (1) Show: If λ_n is the unique eigenvalue of $A \in \mathbb{R}^{n \times n}$ with largest magnitude and x_0 is chosen arbitrarily in $\mathbb{R}^n$, then the iteration $x_k = A^k x_0$ will almost surely converge to a multiple of v_n , the eigenvector associated with λ_n .
- (2) If λ_k is a unique eigenvalue which does not have the largest magnitude from among the eigenvalues of A , then describe a method that will compute the eigenvector corresponding to λ_k .
- (3) Describe the QR iteration algorithm for computing the eigenvalues for a matrix $A \in \mathbb{R}^{n \times n}$. Draw analogues between the QR method and the power method.

5. BACKGROUND: It can be shown that for ϵ small there are differentiable functions $x(\epsilon)$, and $\lambda(\epsilon)$ such that $(A + \epsilon F)x(\epsilon) = \lambda(\epsilon)x(\epsilon)$, where $\|F\|_2 = 1$. Let $x = x(0)$ be a right eigenvector of A , and y be a left eigenvector of A .

- (1) If $A = D + F$, where $A, D, F \in \mathbb{R}^{n \times n}$ and $D = \text{diag}(A)$, then state and prove a theorem relating the location of the eigenvalues of A to the matrices D and F .

- (2) Show: If $A, D, F \in \mathbb{R}^{n \times n}$, $Ax'(0) + Fx = \lambda'(0)x + \lambda x'(0)$.
- (3) Using part (2) above show that $|\lambda'(0)| = \left\| \frac{y^t F x}{y^t x} \right\| \leq \frac{1}{y^t x}$.
- (4) Explain the implications of (3) to the stability of eigenvalues of symmetric matrices, and highly unsymmetric matrices.

6. This problem has five parts.

- (1) Give a careful statement of the error formula for Lagrange interpolation.
- (2) Give a careful proof of the error formula for Lagrange interpolation.
- (3) Give a careful definition of the Chebyshev Polynomials $T_n(x)$.
- (4) Give a careful statement of the theorem that shows that the roots of $T_n(x)$ provide the optimal choice on $[-1, 1]$ for Lagrange interpolation.
- (5) For the function $f(x) = \sin(\pi x)$ defined on $[-1, 1]$ and polynomial interpolating function $p_n(x)$ with interpolating points taken to be the roots of $T_n(x)$, and tolerance $\epsilon = 0.00001$, find an integer n such that the interpolating polynomial $p_n(x)$ has the property that $|p_n(x) - f(x)| < \epsilon$ for all $x \in [-1, 1]$.

7. This problem has two parts.

- (1) Devise a 2nd order method for computing the cube root of a number.
- (2) Show that your method always works.

8. This problem has two parts.

- (1) Explain Romberg's technique for numerical approximation of an integral.
- (2) Discuss the ideas and concepts that explain why the method works.

9. This problem has two parts.

- (1) Set up the system of linear equations to be solved when a finite difference approach is used to solve the two point boundary value problem: $v'' + b(x)v' + c(x)v = d(x)$, where $v(0) = 0$ and $v(1) = 0$.
- (2) Set up the system of linear equations to be solved when a collocation approach (with basis functions $\sin(k\pi x)$) is used to solve the two point boundary value problem: $v'' + b(x)v' + c(x)v = d(x)$, where $v(0) = 0$ and $v(1) = 0$.

10. This problem has three parts.

- (1) Describe the defining properties of the Legendre polynomials on the interval $[-1, 1]$.
- (2) Give a careful statement of the theorem fundamental to Gauss Quadrature.
- (3) Show how the Gauss Quadrature method can be used to approximate the integral

$$\int_0^1 \exp(x^2) dx.$$